Parametric Multimodal User Memory: Storing What Captions Cannot Carry

Bojie Li
Pine AI

Noah Shi
University of Washington

Abstract

A personalized agent needs a *user memory*: a persistent model of who its user is. Today that memory is almost always *text* — transcripts and captions, retrieved by similarity search. This serves the *captionable* half of a person ("my cat is named Bibi") but drops the *perceptual* half that no caption can hold: how a voice sounds, how a face reads across age and lighting, how tired someone sounds today. We measure this loss: across five modalities, a captioner's re-identification note recovers a perceptual identity at as little as 0.11 of a purpose-built encoder's recall, collapsing toward chance where the signal is not nameable.

We instead **ground** perceptual memory in the model, decomposing recall into two subproblems. A vision-language model grounds the referent in context (*what* and *where*); a purpose-built encoder turns that region into an identity *key* (*who*); and the key is stored as one inline token, read by attention inside generation with no round-trip. Neither suffices alone: the VLM identifies cross-age faces at only 0.54, against 0.81 for a face encoder, while the encoder cannot pick a referent from a scene (0.05). Grounded, the system reaches the correct-region oracle (0.96); the same decomposition grounds a speaker in multi-speaker audio and a person in video; and the recognition core is *training-free*, reproducing the encoder's recall on any frozen model. On **PerceptMem** (12 domains, 1,080 tasks) we further find perceptual identity capacity-limited (recall $\approx \min(1, k/M)$) and exact facts binding-limited, belonging in a text store. The two memories compose, preserving text recall exactly: an agent with both remembers what its user said *and* what they were like.

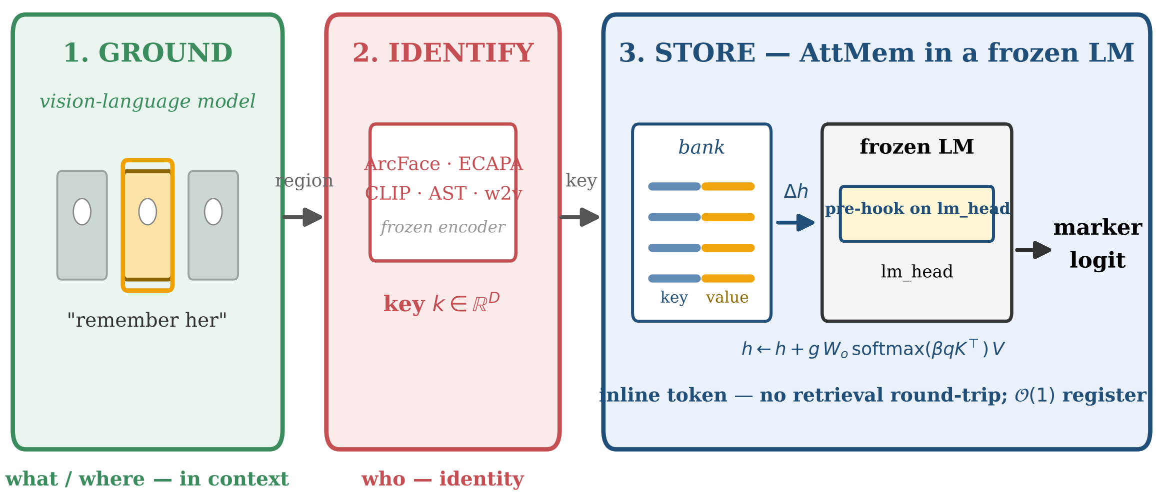


Figure 1. A vision-language model grounds the referent the user means in a cluttered scene (*what*/*where*); a purpose-built encoder turns the grounded region into an identity key (*who*); and AttMem stores the key on a frozen language model, reading it back as a single inline token. Neither component alone clears chance; grounded, they recover the correct-region oracle (Figure 2). The mechanism is detailed in Figure 4.

1 Introduction

When an AI agent "remembers" something about the person it serves, what does it keep? In almost every system today, *text*. Mem0 [Chhikara et al., 2025], MemoryLLM [Wang et al., 2024], MemGPT / Letta [Packer et al., 2023, Team, 2024], M3-Agent [Long et al., 2025], and the wider conversational-memory line [Wu et al., 2024] turn what the agent saw or heard into transcripts (via ASR [Radford et al., 2022]) and captions (via BLIP-2 [Li et al., 2023] or LLaVA [Liu et al., 2023]), store the fragments in a sentence-encoder vector index [Reimers and Gurevych, 2019], and search them later. Personalized vision-language systems (MyVLM [Alaluf et al., 2024], Yo'LLaVA [Nguyen et al., 2024], MC-LLaVA [An et al., 2024], Online-PVLM [Bai et al., 2025], RAP [Hao et al., 2025]) keep the same textual spine: a *named* concept ("Bibi") is the handle, the perception an afterthought.

This is exactly right for one half of a person. "My cat is named Bibi." "My favourite restaurant is in Paris." "I am vegetarian." These are *captionable*: they pass through text intact and come back whole. But they are not the whole person. Consider what else an attentive companion carries about someone:

- How their voice sounds when they say their own name, and how that differs from a stranger saying it.
- How their face reads today: a little older than last year's photo, but unmistakably them.
- How their brushwork looks: "this is by the painter from Tuesday, even though it's from a different period."

None of this survives text: "a brown-haired man" does not tell two brown-haired men apart, and no better caption closes the gap because the loss is a property of the channel, not the wording. We confirm it is measurable (Section 5): a re-identification note from a strong captioner recovers a perceptual identity at as little as 0.11 of a purpose-built encoder's recall. So an agent's memory has two halves. *Captionable* content (facts, preferences) belongs in a text store, where sentence-encoder similarity works. *Perceptual* content — how a face, a voice, or a painting actually *is* — has no home today: it is flattened into a caption and lost, or handed to a recognizer that returns a bare label like "speaker 47" the model never sees. The missing piece is a memory that stores a perception *as a perception*.

Our proposal: grounded, parametric, multimodal user memory. Remembering a perception decomposes into two subproblems (Figure 1). First, *what* to remember and *where* it is: in a real scene the referent the user means (one face among several, the painting being discussed) must be grounded in context, a joint language-and-vision judgement. Second, *who* or *what* it is: the grounded region must be turned into a stable identity that survives changes in lighting, age, room, or recording, which is what a purpose-built perceptual encoder does. We assign each subproblem to the component that can solve it. A vision-language model grounds the referent in context; a dedicated encoder (ArcFace [Deng et al., 2019], ECAPA-TDNN [Desplanques et al., 2020], CLIP [Radford et al., 2021]) turns the grounded region into an identity *key*; and the key is stored as a single row in a small per-modality bank bolted onto a frozen language model, paired with the model's own vector for a marker token as the *value*, and read back by attention at the output head. Recall is then a token the model conditions on, produced inside generation with no caption and no retrieval round-trip. Registering an identity is a single tensor append,

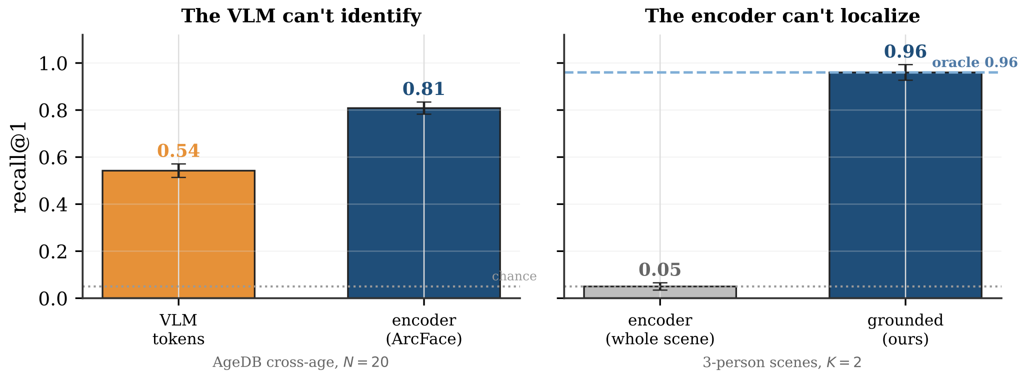


Figure 2. Neither component alone; grounding fixes both blind spots. Two *separate* experiments, each isolating one failure. *Left (identity):* on the same AgeDB cross-age faces ($N=20$), the VLM's own vision tokens are a weak identity encoder (0.54) beside a purpose-built face encoder (0.81). *Right (localization):* in three-person scenes ($K=2$), a bare encoder embedding the whole scene is near chance (0.05), while grounding the referent first recovers the correct-region oracle (0.96). The division of labour is a requirement, not a convenience: each component covers the other's blind spot, and the detailed ablation follows in Figure 7.

with no per-user training; recall is constant-time. We call the in-model bank AttMem (for *attention memory*); it borrows the shape of kNN-LM [Khandelwal et al., 2020] and Memorizing Transformers [Wu et al., 2022] but aims it at a new target, the persistent perceptual identity of a user, registered in $\mathcal{O}(1)$.

Why grounding is necessary. It is tempting to think a perceptual memory is just an encoder run over whatever the agent saw. It is not, because a raw perception mixes the identity worth remembering with the context around it. A friend photographed at a coffee shop and the same friend surfing share almost nothing pixel-for-pixel — background, clothing, pose, and light all change — so a single embedding of the whole image places the two far apart, even though we recognise them at a glance by *looking at the face* and mentally cropping the scene away; only the face, cut out and compared, is invariant. Audio is the same: one voice on a phone call and on a podcast, on different topics, shares few words and different acoustics, yet a listener locks onto the speaker while a whole-clip embedding drifts with content and setting. **Grounding is this act of cutting out the referent before encoding it**, and it is what makes the encoder's invariance usable in the wild. It also forces a division of labour, because neither component can do it alone (Figure 2): a vision-language model can resolve *which* region to cut but is a weak identity encoder, while a specialist encoder is invariant to condition but cannot pick the referent out of a cluttered scene. Grounded, each covers the other's blind spot, and the system recovers oracle-level recognition that neither reaches alone.

Figure 3 contrasts the two routes on one "remember her" episode.

Contributions.

- **Perceptual user memory must be grounded, not captioned.** We argue, and measure, that text-shaped memory cannot carry perceptual user content (a text note recovers a perceptual identity at as little as 0.11 of an encoder's recall), and that recall must instead be decomposed into VLM grounding, encoder identity, and an in-model store (Sections 1–3).

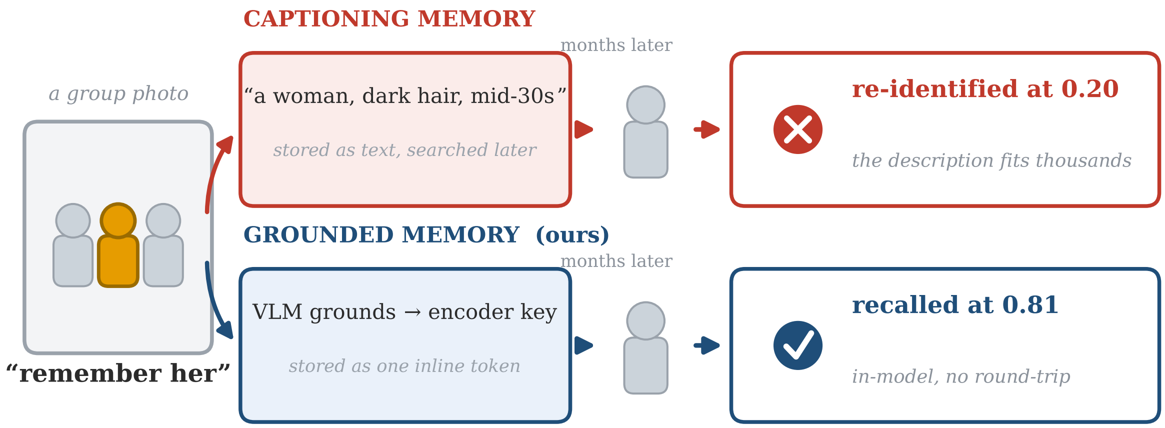


Figure 3. A worked example. "Remember her" in a group photo: a captioning memory stores a description that fits thousands and re-identifies her near chance later, while the grounded memory stores one inline token and recalls her cross-condition inside the model. Numbers are paired re-identification accuracies (Section 5).

- **A grounded memory that recovers oracle recognition in context.** The VLM grounds the referent, a purpose-built encoder identifies it, and the identity is stored as an inline attention token. In cluttered scenes this recovers the correct-region oracle (0.96 on faces, 0.36 on paintings, both equal to oracle) where whole-scene encoding is near chance (0.05), across two visual domains and two encoders (Section 5).
- **A training-free, $\mathcal{O}(1)$ in-model recognition core.** The store is read by attention at the output head with *zero* gradient steps; it reproduces the encoder's recall exactly on ten frozen model families (1.5B–14B, tied and untied embeddings, transformer and hybrid-Mamba), with one read-strength constant as the only knob. Insertion is $\mathcal{O}(1)$ and recall is constant ($\sim$15 ms), 52× faster than feeding the same memory in as context, which runs out of memory entirely past ten thousand identities (Sections 3–5).
- **PerceptMem and a capacity-law router.** A benchmark of 12 perceptual domains and 1,080 tasks across five modalities (face, painter style, speaker, acoustic scene, tone of voice), with text-only and chance baselines on every comparison, plus two capacity laws that decide what belongs in the perceptual bank versus a text store: perceptual identity is capacity-limited (recall $\approx \min(1, k/M)$), exact facts are binding-limited (Sections 4, 7).

Code. Our implementation and the PerceptMem benchmark are available at `https://github.com/19PINE-AI/multimodal-user-memory`.

2 Decomposing perceptual recall: grounding and identification

Why does the text route fail, not for lack of effort but structurally? Take the voice example in its hardest form: ten samples on file, a new one arriving cross-recording across different rooms and days, and the agent must name which of the ten it is. An agent has four options (Table 1). Three discard the very signal that tells one voice from another. The fourth keeps it — and exposes a second issue every route quietly conflates: recognizing a perception is really *two* problems, localizing the referent in context and encoding its identity, and no single component does both well.

Table 1. Four routes for remembering a perception, plus ours (citations in text). Only routes that keep the raw perception can discriminate across conditions. Embedding retrieval and ours both inherit the encoder's recall exactly; ours additionally localizes the referent with a VLM and reads the identity back *inside* the model as a token the model conditions on, training-free and in $\mathcal{O}(1)$.

Route	What it stores	Keeps percept?	Cross-condition	Register
Caption-and-search	text description	no	chance	$\mathcal{O}(1)$
Recognize-and-label	a bare label	no	label only	$\mathcal{O}(1)$
Per-concept tuning	tuned token/adapter	yes	strong	grad./id
Embedding retrieval	raw embedding	yes	encoder-limited	$\mathcal{O}(1)$
Parametric (ours)	embedding + LM value	yes	= **encoder, in-model**	$\mathcal{O}(1)$

The three that discard the signal. *Caption-and-search* [Chhikara et al., 2025, Wang et al., 2024] writes the voice down ("a male voice, mid-30s, slight Eastern-European accent") and retrieves by similarity over the words. The description is true and useless. It fits thousands of speakers, so cosine over captions cannot separate ten users with even coin-flip reliability. *Recognize-and-label* [Long et al., 2025] calls a recognizer (ArcFace [Deng et al., 2019], ECAPA-TDNN [Desplanques et al., 2020]) and stores the label it returns ("speaker 47"). The model can recall that a label was assigned, but it never holds the perception and cannot compare today's voice against last week's. *Per-concept tuning* [Nguyen et al., 2024, Alaluf et al., 2024] fine-tunes a token or adapter per concept. It works, but each identity costs a second of gradient descent and yields a concept-specific artifact — the wrong cost profile for a companion registering dozens of characteristics per user, continuously.

The one that keeps it, and what it still needs. *Embedding retrieval* skips the words, indexes the raw voiceprint, and retrieves by cosine over the encoder's vectors. It is the only text-free route that keeps the perception, and the strongest baseline we take seriously throughout. It succeeds exactly to the extent that the encoder is invariant across recording conditions, and fails to the extent that it is not. On a 2180-identity face pool it reaches 0.93 recall at ten identities — strong, but short of certainty, and eroding as the pool grows. Crucially, embedding retrieval presumes the perception has *already* been isolated: someone handed it a clean voiceprint or a tight face crop. In a real scene that isolation is itself a hard, context-dependent step, and it is where a bare encoder is helpless (a single embedding of a three-person photo recognizes the intended person at 0.05, Section 5). This is the second problem the four routes conflate: *localizing* the referent is not the same as *identifying* it, and the strongest text-free route only solves the second.

Our mechanism keeps the encoder's identity step and adds the two things it lacks. It puts a VLM in front to localize the referent the user means, and it changes where the answer lives: it stores the voiceprint as a key but pairs it with the language model's *own* internal vector for a marker token as the value, and reads the pair back by attention inside the model. The recalled value is then a token the model conditions on, not a label handed back from an external index. The recognition is identical to retrieval's — with a sharp read the attention is a hard argmax over the same encoder cosine, so the memory reproduces embedding retrieval exactly, never beating it. The contribution is the grounding around that core: the referent is resolved in context, the identity survives conditions text cannot, the recall lives inside the model, registration is $\mathcal{O}(1)$, and the mechanism works without modification on any frozen model.

3 Grounding perceptions in the model's own representation

Recall decomposes into three steps, and we take each in turn: *ground* the referent in context (the VLM), *identify* it with a dedicated encoder, and *store* the identity as a token read inside a frozen model. The first two are off-the-shelf components placed where they are strong; the third, AttMem, is the mechanism this paper contributes.

Grounding: resolving the referent in context. A perception rarely arrives pre-cropped. The user gestures at one person in a group, names the painting on the left, refers to the voice that just spoke; the system must turn that context-dependent reference into a region before anything can be encoded. This is exactly what a vision-language model is good at and an encoder is not: the VLM grounds the referent against the surrounding scene and the user's words, returning a region (a bounding box, a span of audio). We use the VLM's native grounding for this and re-detect inside the returned region where a modality detector exists (RetinaFace for faces, from the same InsightFace project as ArcFace), so box imprecision and un-aligned scenes are handled before identity is read. The necessity of this step is empirical: encoding the whole scene, with no grounding, recognizes the intended referent at 0.05 on faces and 0.11 on paintings, near chance, because a single embedding cannot represent a chosen referent among distractors (Section 5).

Identification: a dedicated encoder, not the model's own tokens. The grounded region is read by a dedicated perceptual encoder into an L2-normalized identity *key*: ArcFace for faces, ECAPA-TDNN for speakers, CLIP for painting style, AST [Gong et al., 2021] for acoustic scenes, a wav2vec2 emotion encoder [Baevski et al., 2020] for tone. It is tempting to skip this and read identity from the VLM's own context-conditioned representation — to have the model "output the embedding of the face part" directly. We tested that and it does not hold: a linear probe on a context-conditioned VLM hidden state recovers the referent at 0.25 (raw hidden state 0.14, against an oracle of 0.80), and the VLM's native vision tokens recognize AgeDB cross-age faces at 0.54 where ArcFace reaches 0.81 at the same $N{=}20$ (Figures 10, 6); the gap persists end-to-end inside a real VLM (§6, Appendix Table 10). The VLM is a strong localizer and a weak identity encoder; the division of labour is not a convenience but a requirement. The key must be a dedicated perceptual encoder, separate from the model's own representation, which is also why the mechanism pairs naturally with an omni model's encoder used as an external feature rather than its in-context tokens.

Storage: an in-model attention memory. The identity key is stored and recalled by AttMem, a table of rows on a frozen language model (Figure 4). Row i holds a *key* $\boldsymbol{k}_i$ (the encoder embedding above) and a *value* $\boldsymbol{v}_i$ (the language model's own embedding for a marker token assigned to that identity). Storing the model's native vector as the value, rather than an arbitrary learned one, is deliberate: it makes the eventual logit boost for the right marker clean and self-reinforcing, rather than an accident of two unrelated vectors.

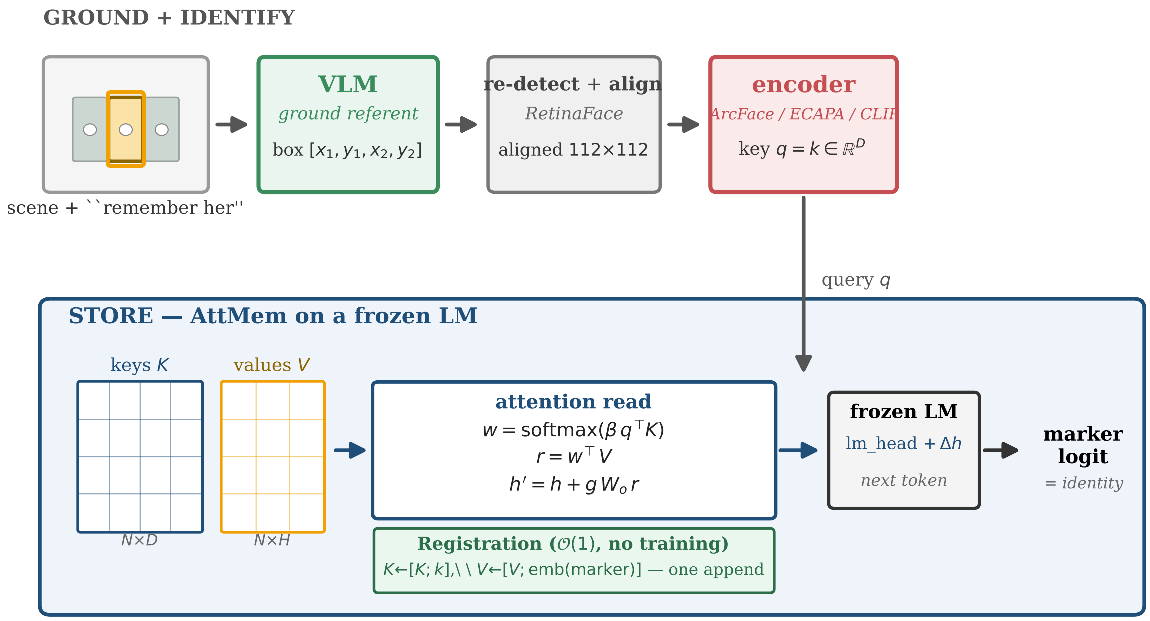


Figure 4. The grounded memory in detail. **Ground + identify (top):** the VLM resolves the referent to a box, a modality detector re-detects and aligns the region (RetinaFace for faces), and a frozen perceptual encoder (ArcFace / ECAPA / CLIP) maps it to an L2-normalized key $q{=}k \in \mathbb{R}^D$. **Store (bottom):** AttMem holds keys K ($N{\times}D$) and values V ($N{\times}H$, the model's own marker-token embeddings); the attention read $\boldsymbol{w}{=}\mathrm{softmax}(\beta\, \boldsymbol{q}^\top K)$, $\boldsymbol{r}{=}\boldsymbol{w}^\top V$, $\boldsymbol{h}'{=}\boldsymbol{h}{+}g\, W_o \boldsymbol{r}$ adds a residual at the output head that biases the next token toward the matching marker. The read has no trainable parameters (β, g are constants); registration is a single $\mathcal{O}(1)$ append.

Reading the memory is one step of attention. When a localized perception arrives at generation time, the model forms a query $\boldsymbol{q}$ from its encoder key and, just before the output head, computes

$$\boldsymbol{w} = \mathrm{softmax}\big(\beta\, \boldsymbol{q}^\top \boldsymbol{K}\big), \qquad \boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V}, \qquad \boldsymbol{h}' = \boldsymbol{h} + g \cdot W_o\, \boldsymbol{r},$$

where $\boldsymbol{K}, \boldsymbol{V}$ stack the bank's keys and values, β is the attention sharpness, g a residual gain, and W_o the identity. In words: the perception softly looks up the bank, pulls back a blend of the matching markers' model-side vectors, and nudges the next-token prediction toward them. The blend is the entire read, and it adds *no trainable parameters*: β and g are two constants, and the values are the model's own marker embeddings.

Design choices for a faithful read. Four details determine whether the read recovers the encoder's nearest neighbour cleanly, and getting them right is what lets the mechanism work with no training. (i) *Do not* divide the attention logits by $\sqrt{D}$: with normalized keys, that flattens every cosine difference into near-uniform attention. (ii) Set the sharpness β high, so the attention is a near-hard argmax over the bank rather than a diffuse average. (iii) Attach the hook at the output head, not at an intermediate layer, so the residual reaches the logits undiluted. (iv) Set the gain g large, so the retrieved marker dominates the hidden state and the read is exact; on models whose output embeddings are small in norm, g must be larger

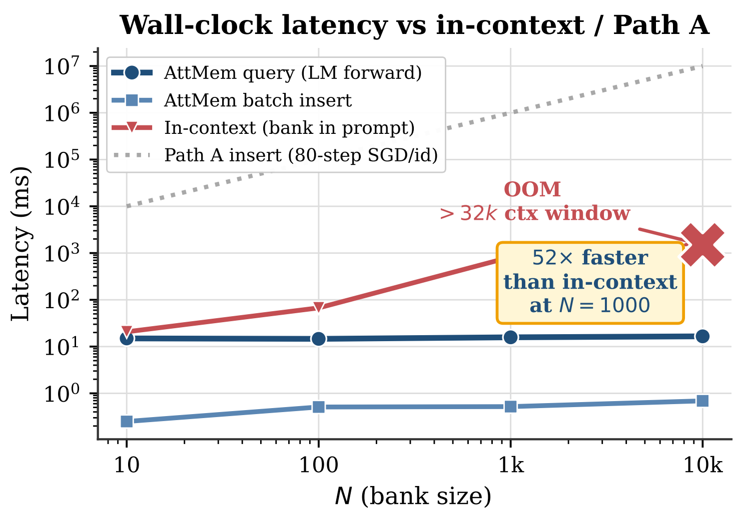


Figure 5. Recall latency is flat in bank size ($\sim$15 ms, dominated by the frozen model's own forward pass) and insertion is $\mathcal{O}(1)$. Feeding the same registered perceptions in as context grows linearly and exhausts the context window past ten thousand identities. Per-concept fine-tuning (PATH A-style) costs gradient steps per identity. Log–log axes.

still (Section 6). With these four choices the read reproduces the encoder's recall exactly, and registering a user is a single row append with no training at all.

Registration versus fine-tuning. The cost of writing to memory is a decisive property (Figure 5). Adding an identity is a single `torch.cat` (about half a millisecond to add a thousand), and recall is constant-time regardless of bank size, because the lookup is a small matrix multiply dwarfed by the model's own forward pass ($\sim$15 ms either way). The natural alternative, feeding the registered perceptions into the model as context, grows linearly per query and exhausts memory past the context window. Our mechanism is 52× faster at a thousand identities, and is the only one of the two that still functions at ten thousand. A memory that a companion updates after every conversation must be this inexpensive to write.

4 PerceptMem: a cross-condition perceptual-memory benchmark

To measure perceptual recall we built **PerceptMem**, five sub-modalities each chosen for one reason: the signal that discriminates is provably destroyed by captioning (Table 2). They span vision and audio, identity and style, the physical and the affective:

Table 2. The five PerceptMem sub-modalities. Each is included because no caption can carry its discriminating signal.

Sub-modality	Source / encoder	Why text cannot carry it
Face across age & lighting	LFW+AgeDB / ArcFace	"brown-haired man" fits thousands
Painter style	WikiArt / CLIP-mid	no caption separates early vs. late Monet
Speaker across recordings	LibriSpeech / ECAPA-TDNN	timbre is not transcribable
Acoustic scene	ESC-50 / AST	"traffic noise" fits every street
Tone of voice	RAVDESS / wav2vec2	"sounded tired" lacks a personal baseline

The sources are all standard: LFW [Huang et al., 2008] and AgeDB [Moschoglou et al., 2017] read with ArcFace [Deng et al., 2019], WikiArt [WikiArt, 2010] with mid-layer CLIP [Radford et al., 2021], LibriSpeech [Panayotov et al., 2015] with ECAPA-TDNN [Desplanques et al., 2020], ESC-50 [Piczak, 2015] with AST [Gong et al., 2021], and RAVDESS [Livingstone and Russo, 2018] with a wav2vec2 emotion encoder [Baevski et al., 2020]. The benchmark deliberately tests *memory*, not perception: every sub-modality has a fixed, off-the-shelf encoder that any method may use as black-box infrastructure, so no one wins by having a better encoder. The interface is just `register` and `recall`. Data is cross-session: a registered sample and its query come from different recordings. The identities used to train the memory never overlap with those it is evaluated on. For a memory of N identities we register one sample each, then issue cross-condition queries and ask the method to name the right one. For tone of voice, each registered "identity" is a (speaker, emotional-state) pair, so recalling it means matching a paralinguistic state across separate utterances, something a per-utterance caption cannot do without the user's own history.

For statistical rigor we expand these five conceptual sub-modalities into a 12-domain, 1,080-task benchmark across multiple datasets and encoders (Section 5, Figure 10), and report a text-only baseline and a chance floor on every comparison. Throughout, our reference point is **embedding retrieval**: the same encoder, cosine nearest-neighbour over the registered keys. We call it the *encoder ceiling*, because it is the best one can do with the encoder's own similarity, and our in-model read reproduces it rather than exceeding it.

5 Empirical evaluation

We organize the evaluation around the claims of the grounded memory. First, the half text drops is real and large: captions cannot carry perceptual identity (§5.1). Second, grounding recovers it: VLM grounding plus an encoder identity reaches the correct-region oracle in cluttered scenes (§5.2), and the same decomposition generalizes to audio and video (§5.3). Third, the recognition core is the encoder, read in-model, which is a feature, not a shortfall (§5.4). We then show that the in-model store enables composition with text memory (§5.5) at a cost no external index can match (§5.6) while preserving text behaviour exactly (§5.7), and assemble the pieces into an end-to-end multi-session agent (§5.8).

5.1 Text captions cannot carry perceptual identity

The baseline a system would otherwise use is a text note: describe the perception in words, store the sentence, retrieve by text similarity. We test this on *all five modalities* and give it its best shot — an LLM writes a re-identification note (Qwen2.5-VL for images, Qwen2.5-Omni for audio), a sentence encoder embeds it, and recognition is cosine nearest-neighbour over the notes — scored against the encoder on identical draws. The result is a clean router gradient (Figure 6): the note's recall slides from nearly the encoder's where the signal is *nameable* (an acoustic scene, "someone typing on a keyboard") down toward chance where it is a perceptual *identity* (a voice, whose "high-pitched, with an accent" fits countless speakers). Text is a competent memory for what can be said and a failing one for what cannot — precisely the line the router draws between the text store and the parametric bank (§7).

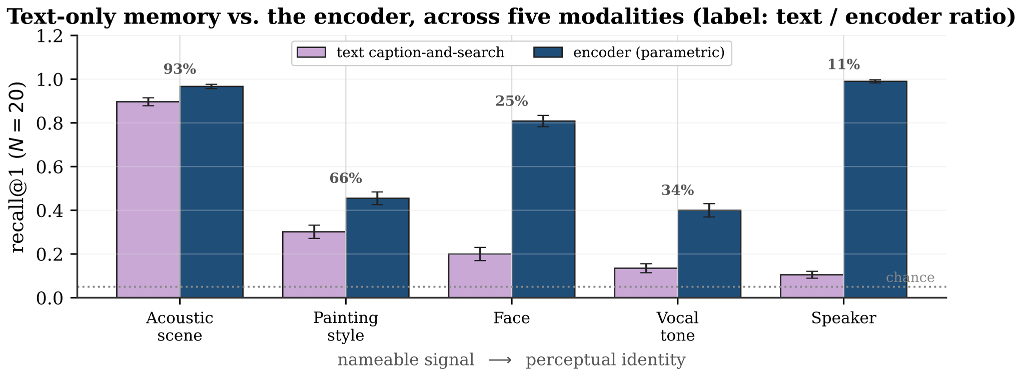


Figure 6. Text-only caption-and-search versus the parametric encoder, paired on identical draws across all five modalities (N=20, 30 draws, 95% CI); the label above each pair is the text/encoder ratio. Ordered by how nameable the signal is. Text nearly matches the encoder on the nameable acoustic scene (93%) and collapses toward chance on perceptual identity (speaker 11%), tracing the router's boundary between the text store and the parametric bank.

5.2 Grounding recovers oracle recognition in cluttered scenes

In deployment a perception rarely arrives pre-cropped: the referent sits among other people or objects, and the system must resolve the one the user means before encoding it. An ablation isolates the value of each step (Figure 7). A *text-only* memory, in which the VLM describes the referent from the whole scene and captions are matched, recovers it at 0.12 on faces and 0.17 on paintings; a *store-only* memory that embeds the whole scene is lower still, near chance, because a single embedding cannot represent a chosen referent among distractors. *Grounding* the referent first, then identifying it with the encoder, recovers the correct-region oracle (0.96 on faces, 0.36 on paintings), with the VLM resolving the right region on essentially every trial. Retrieval coincides with the store throughout — both are the encoder's nearest neighbour — so the gain is grounding, not the matcher.

The advantage is robust to scene clutter and VLM scale (Figure 8). As the number of referents per scene grows from two to six, grounding accuracy stays at 1.00 through K=5 and slips only marginally at K=6 (0.99), with grounded recall tracking the oracle (0.95–1.00) throughout while whole-scene recall stays at the floor; text-only, by contrast, decays as clutter grows, halving from 0.12 at K=2 to 0.05 at K=3 as the caption must disambiguate more distractors. Grounding accuracy is already saturated at 7B and unchanged at 32B (1.00 either way, recall 0.960 vs 0.958), so the decomposition does not depend on a large grounder. Grounding is lossless and domain-general; what it recovers is bounded only by the encoder, modest for CLIP style and near-perfect for ArcFace identity. This is the decomposition made visible: the VLM resolves *what* to remember and where it is, and a purpose-built encoder determines *who* it is.

5.3 Grounding generalizes to audio and video

The decomposition is not vision-specific: the same three steps apply when the referent is a *speaker* in a multi-speaker recording or a *person* in a video (Figure 9). For **pure audio**, we concatenate K real VoxCeleb speakers into one clip and ask an audio-LLM (Qwen2.5-Omni) to ground the referenced speaker's time span — the direct analog of the VLM grounding a bounding box — then identify that span with ECAPA. Grounded recall equals the oracle segment at K=2 (1.00) and K=3 (0.92, grounding 0.96) and degrades at K=4 (0.67, grounding

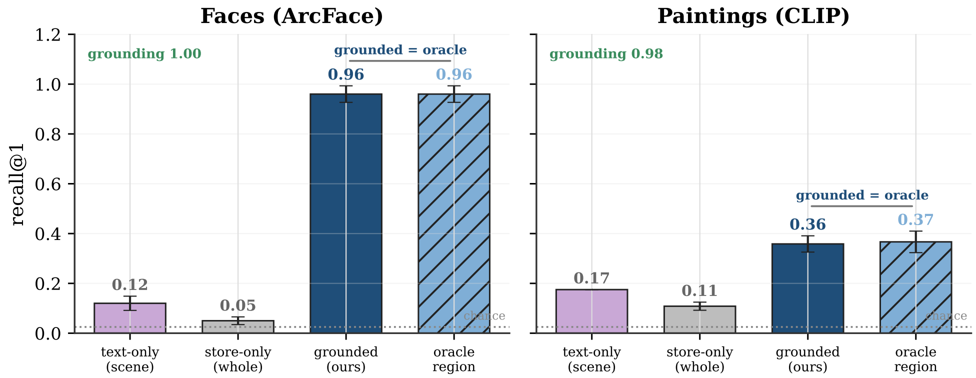


Figure 7. Ablation of the grounded memory on cluttered scenes (M=40, two referents per scene). Capability increases left to right: *text-only*, where the VLM describes the referent from the whole scene and captions are matched; *store-only*, embedding the whole scene with no grounding; *grounded* (ground + identify + store, ours); and the correct-region *oracle*. Both text-only and store-only are near chance, while grounding recovers the oracle. Retrieval coincides with store throughout (both are the encoder's nearest neighbour), so the gain is grounding, not the matcher. Faces: ArcFace, landmark alignment, 5 seeds; paintings: CLIP, no alignment, 3 seeds; error bars ±95% CI.

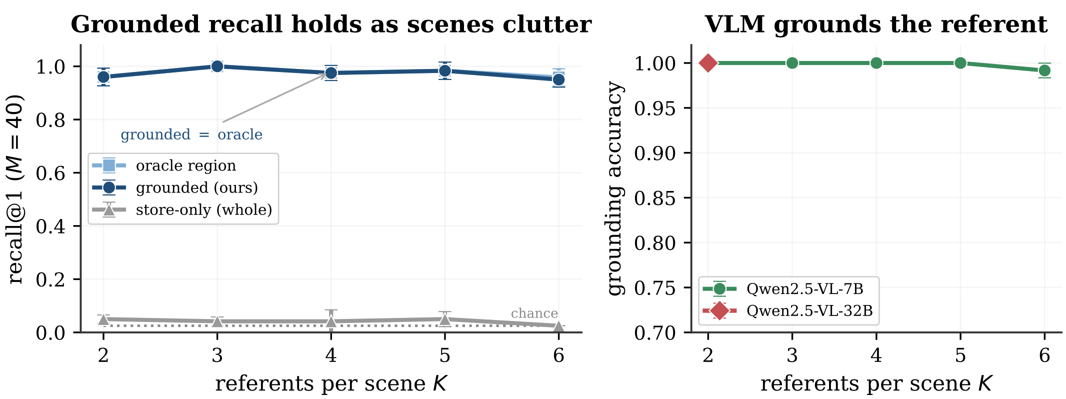


Figure 8. The grounded advantage is robust to scene clutter and grounder scale (faces, M=40). **Left:** as referents per scene grow from two to six, grounded recall (navy) stays at the correct-region oracle (light blue, coincident) while whole-scene recall stays at the floor. **Right:** grounding accuracy stays at 1.00 through K=5 and dips only to 0.99 at K=6, and is identical for Qwen2.5-VL-7B and -32B, so the decomposition does not require a large grounder.

0.69) as temporal grounding of the fourth speaker becomes hard, while whole-clip encoding stays near chance and the encoder itself is always perfect. The result carries to real conversation: on *VoxConverse*, in-the-wild YouTube conversations with natural turn-taking, overlap, and noise, grounding is harder (0.75) but the grounded read (0.73) still nearly doubles whole-window encoding (0.38) toward the oracle (0.97), across 60 conversations, and the *AMI* meeting corpus behaves the same (grounded 0.75 vs whole-window 0.34, grounding 0.88). This mirrors the vision finding exactly: grounding, not identity, is the component that degrades with complexity.

For **video** (vision and audio together), we use real RAVDESS recordings, in which the same 24 actors are captured on video across eight emotions, giving a genuine (face, voice) pair per identity. We register each actor from one clip (an ArcFace face key and an ECAPA voice key)

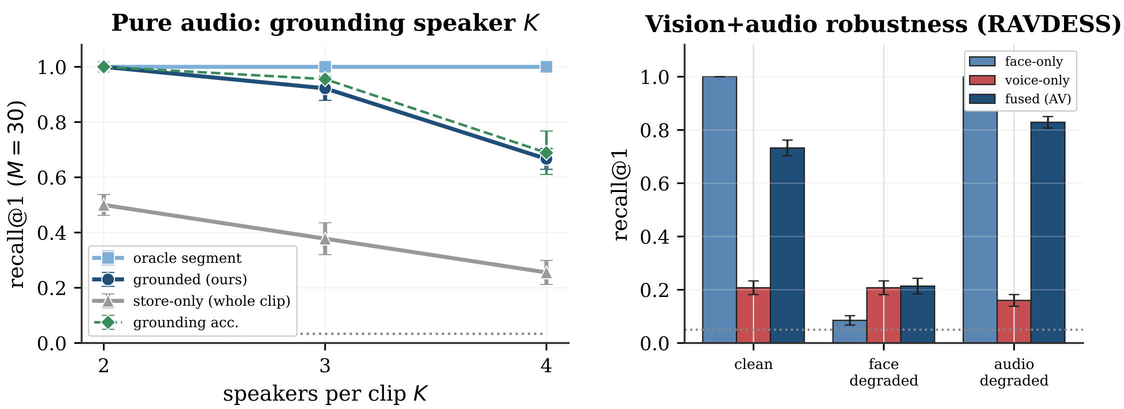


Figure 9. Grounding generalizes beyond vision. **Left — pure audio:** an audio-LLM grounds the K-th speaker's time span in a multi-speaker VoxCeleb clip and ECAPA identifies it (M=30, 3 seeds). Grounded recall (navy) tracks the oracle segment at K=2–3 and degrades at K=4 as temporal grounding gets hard; whole-clip encoding stays near the floor. **Right — vision+audio (RAVDESS video):** each identity carries a face and a voice key. Storing both is robust to single-channel corruption: face-only collapses when the face is a distant low-resolution view, voice-only is weak and drops under noise, but the fused memory never falls below the surviving channel (M=20, 40 draws).

and recognise a held-out clip of a different emotion. Here face recognition is near-saturated (1.00) and cross-emotion voice recognition is hard (0.21), so fusing the two channels with equal weight does not beat the face alone — and we do not claim it does. The value of storing *both* is robustness to a corrupted channel, which a single-modality memory cannot provide (Figure 9, right). When the face is degraded to a distant, low-resolution view, face-only recall collapses to near chance (0.09) but the fused memory tracks the surviving voice (0.21); when the audio is corrupted by noise, voice-only drops (0.16) but the fused memory tracks the surviving face (0.83). Fusion is the only representation that never collapses across conditions, which is precisely the condition a deployed companion faces as people turn off-camera or speak over noise.

5.4 The in-model read reproduces the encoder's recall

Once the referent is localized and encoded, recognition is the bank's job, and the bank reproduces the encoder. With a sharp read the attention is a hard argmax over the same encoder cosine that embedding retrieval uses, so its recall equals the encoder's, neither better nor worse. A paired protocol that scores both on identical registrations and queries with the target slot randomised confirms it: the memory ties retrieval on random banks and trails it only on banks of near-identical look-alikes the encoder itself conflates, and a learned metric on the same features lands in the same place (Appendix A). The value of the in-model read is not a higher number but its location: the result is a token inside the model's forward pass rather than a label from an external index.

Breadth and statistical rigor. At scale the recognition core holds across 12 domains and five modalities — 1,080 tasks ($N \in \{10, 20, 40\}$, 30 draws, 95% CIs), clearing chance in every one (Figure 10). Recall *tracks the encoder*: near-perfect where the encoder is strong (speaker, face), low where it is weak (style, tone), and, tellingly, the VLM's own native vision encoder reads

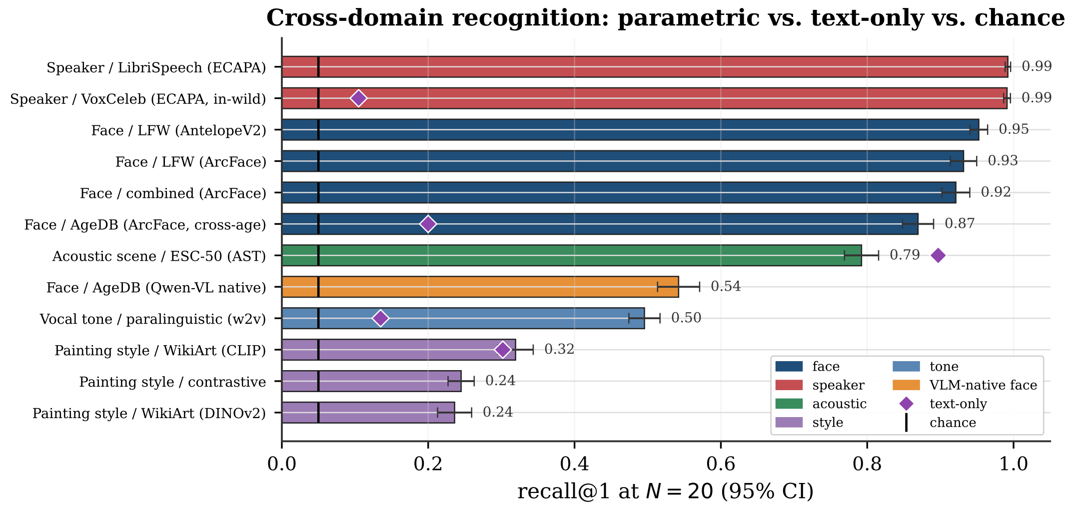


Figure 10. The recognition core across 12 perceptual domains and 5 modalities (1,080 tasks; recall@1 at N=20, 30 draws, 95% CI), coloured by modality, against chance ($1/N$, black ticks) and the text-only caption baseline where measured (purple diamonds, per modality). Parametric recall tracks the encoder and clears chance everywhere, far exceeding text-only on perceptual identity (speaker, face, tone). The one inversion is the *nameable* acoustic scene, where a caption ("typing on a keyboard") edges the weaker AST encoder — exactly the router boundary (§7): text suffices for nameable categories, the parametric bank for identity. The VLM's own native vision encoder (orange) reads the same AgeDB faces far below a purpose-built one.

the same faces far below a purpose-built one — the cleanest statement that the encoder, not the memory, sets each ceiling (§6).

Training-free on any frozen model. Because the read is a similarity-weighted blend of marker tokens, nothing about the recall depends on the host language model, and it ports without training. Across ten families (Qwen2.5 1.5B/7B, Qwen3 4B/8B/14B, Phi-3.5, SmolLM2, DeepSeek-Llama-8B, Mistral-7B, and a hybrid-Mamba Granite) the read reproduces the encoder's recall (Appendix Figure 16), and a five-architecture by five-modality grid holds to within one point in every cell (Appendix Table 5). On tied-embedding models the match is *exact on every draw*; untied models reach exact recall with one larger read-strength constant; the hybrid-Mamba reader is within a point. This universality is a supporting property of the store, not the headline: it says the "store" step is free to host on whatever model the agent already runs.

Rejecting strangers and verifying identities. A user memory must do more than closed-set recall, in which the query is always one of the registered identities. It must reject a stranger it has never enrolled, rather than assign them a registered name, and decide whether two perceptions are the same person. Both fall out of the same read (cosine over the encoder keys), and we report them with standard operating-point metrics over 30 draws (Figure 11). On verification the memory separates same- from different-identity pairs with AUROC 0.99 on voice and 0.97–0.99 on face (equal-error rate 1.3% and 5–8%); on open-set identification it rejects never-enrolled strangers at 0.96–0.99 while still identifying the enrolled. As everywhere, these track the encoder: they are high where it is strong (voice, face) and low where it is weak

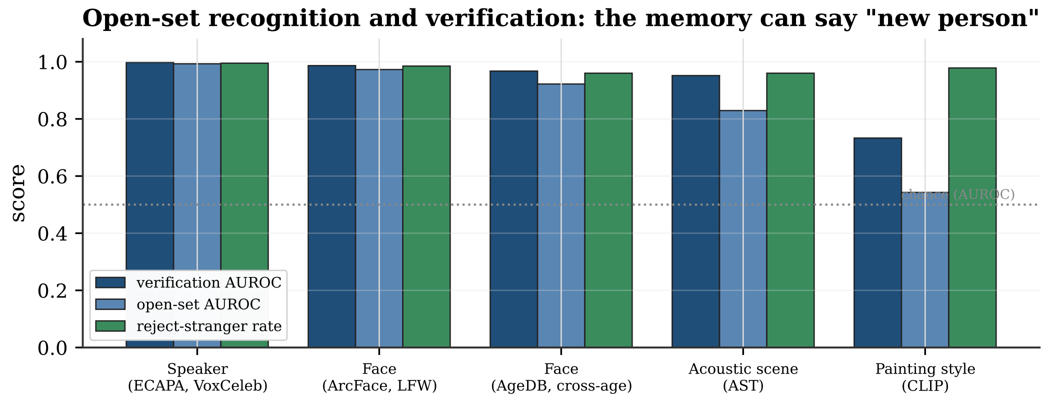


Figure 11. Beyond closed-set recall, the memory verifies identities and rejects strangers (N=20 enrolled, 30 draws). Per modality: verification AUROC (same vs different identity), open-set AUROC (enrolled vs never-enrolled probe), and the reject-stranger rate at the balanced threshold. All are high where the encoder is strong (voice, face) and fall where it is weak (painting style), the same dependence as recall.

(painting style, open-set AUROC 0.54). The point is not a new number but that the memory supports the open-set and verification operating modes a deployed companion requires, inheriting the encoder's separability for both.

5.5 Composition with in-context text memory

The reason to recall a perception as a token, rather than a label from an index, is that the token participates in the model's reasoning. We register M faces with name markers, bind one fact per name in context ("Anna is a teacher"), then show a held-out cross-condition photo and ask for the fact. The whole chain runs in one forward pass — the face recalls the name through the bank, the name retrieves the fact from context — and end-to-end accuracy beats the face-withheld chance baseline by 4–10×, tracking recognition times the model's in-context lookup (Figure 13). What bounds the chain as M grows is that in-context lookup, not the memory. The perception is already a token the model conditions on, folded into the forward pass the agent is running anyway — something an external index, which returns a label outside the model, cannot provide. Figure 12 opens up a single read: a sharp temperature concentrates the encoder's speckled cosine into a near-one-hot diagonal, so the marker logits put the argmax on the same identity retrieval would pick, as a decisive token rather than a diffuse blend.

5.6 Comparison with retrieval and in-context baselines

The natural baseline is retrieve-and-reprompt: cosine-retrieve the nearest registered face, recover its name, and re-prompt the model. Because both routes recognise with the same encoder cosine and reason with the same frozen model, they reach *identical accuracy* — on composition and on a multi-perception "how many distinct people are here?" task alike (Table 3). On cost the pipeline is in fact *cheaper* for pure recognition, since counting distinct identities needs only cosine and no model forward at all. Against this strong baseline the parametric memory is not a faster or more accurate recogniser. Its advantage is architectural: recognition folds into the generation the agent is already running, with no external index, no second prompt, and $\mathcal{O}(1)$ registration, so a single frozen model suffices.

The comparison sharpens against the baseline a VLM actually reaches for: keeping the

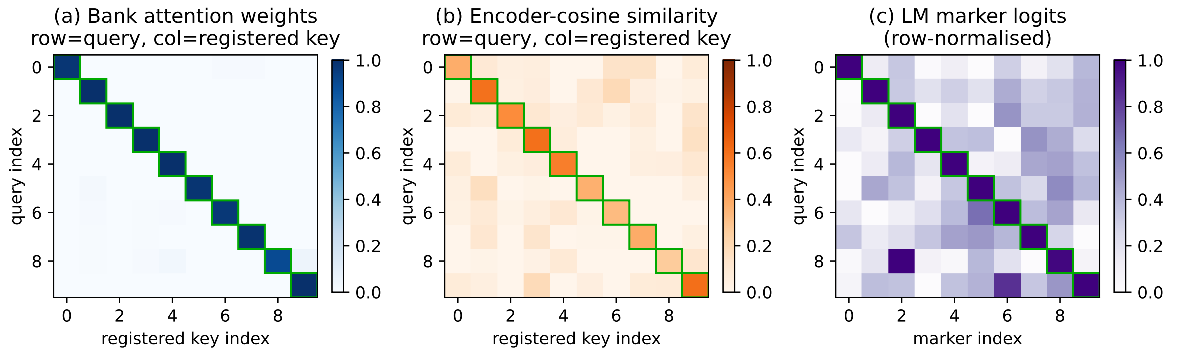


Figure 12. One recall, three views (held-out faces, ten identities). **(a)** with a sharp temperature, bank attention is a near-pure diagonal (avg. 0.98), committing to the top encoder match. **(b)** the underlying encoder cosine has a softer diagonal (avg. 0.46) with off-diagonal mass; the memory inherits whatever errors live here. **(c)** the resulting marker logits keep the argmax identity on top. The attention faithfully reads the encoder's nearest neighbour as a token; it does not sharpen the similarity.

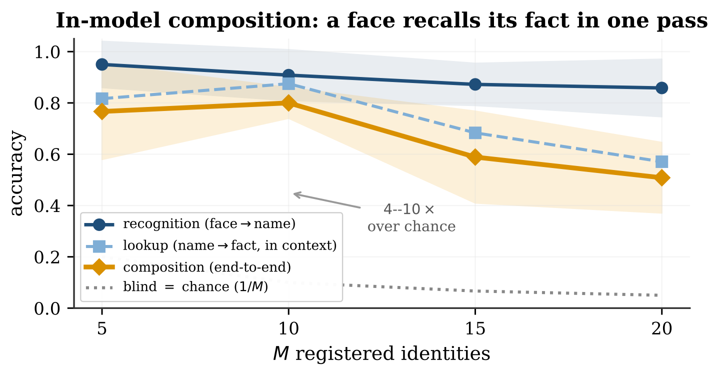


Figure 13. In-model composition: register M faces with name markers, bind one fact per name in context, then recall the fact from a held-out cross-condition photo, all in one forward pass. End-to-end accuracy (gold) tracks recognition (navy) times the model's in-context lookup (blue) and beats the face-withheld chance baseline by 4–10$\times$. 12 draws; bands are $\pm$std.

perception in context as image tokens. A face crop costs 64 vision tokens in Qwen2.5-VL at 224px and 256–576 at higher resolutions (a full image approaches a thousand), where AttMem reads the same perception as *one* marker token. That is a 64–576$\times$ reduction in prefill tokens per perception, and because the bank lives outside the context the memory scales without bound while image-in-context exhausts a 128k window at about two thousand crops. On accuracy this baseline is also far weaker, since the VLM recognises with its general vision tokens (0.60 end-to-end, §6) rather than a face encoder (0.93). So the parametric memory ties the external-index pipeline and outperforms the in-context-image baseline on both accuracy and cost. That, together with the capacity-law router, is the contribution, not a benchmark number.

5.7 Preservation of text-only behaviour

The perceptual memory is meant to sit beside an ordinary text memory, not perturb it. With the bank installed and populated, the model's next-token predictions on text-only prompts are *byte-for-byte identical* to the untouched model (top-1 match on every probe), because the

Table 3. In-model memory vs. a retrieve-and-reprompt pipeline (Qwen2.5-3B, 10 draws). Accuracy is identical everywhere, because both recognise with the same encoder cosine. The pipeline is cheaper for pure recognition (the distinct-count task needs no model forward); the in-model read folds recognition into the model's forward instead of an external index.

Task	In-model acc	Pipeline acc	In-model lat.	Pipeline lat.
Compose (face→fact)	0.81	0.81	122 ms	98 ms
Count distinct, K=2	1.00	1.00	238 ms	0.09 ms
Count distinct, K=3	1.00	1.00	361 ms	0.09 ms
Count distinct, K=4	0.70	0.70	472 ms	0.09 ms

hook adds a residual only at perceptual positions. Register faces and voices together and the two banks stay independent without any training (cross-modal leakage 0.017 and 0.067). An agent can keep "my favourite restaurant is in Paris" in its text store and "how Bibi looks across lighting and age" in its perceptual bank, and lose nothing on either. The discrete-codebook design we tried before this one, which snapped each perception to one of K learned codes, never escaped $\sim$0.07 recall past a few hundred identities for the same reason captioning fails — a categorical bottleneck discards the signal the encoder preserved; we keep it as a contrast in Appendix C.

5.8 An end-to-end multi-session agent

The pieces compose into a working agent. We build one that runs the full system — a frozen language model with the AttMem perceptual bank and a text store of facts — and enroll a growing population of users across sessions, each with a face (registered from one photo) and a bound fact. A returning user is perceived cross-condition (a different photo), and the agent must identify them, answer a question about them, and reject anyone it has never met. We evaluate as the enrolled population grows from ten to eighty users, over five seeds (Figure 14). The perceptual capabilities scale gracefully: identification holds at 0.86 and stranger rejection at 0.81 at eighty users, tracking the encoder as everywhere. Fact recall exposes the router directly. Routed through the text store — the perception recalls the name, the name keys an exact fact lookup — accuracy tracks identification (0.86 at eighty users). Attempted instead *in-model*, by binding all facts in the prompt and reading the answer in one pass, it collapses beyond a few dozen users (0.55 at twenty-five, 0.04 at fifty), because a 3B model cannot hold fifty name–fact bindings in context. This is the router thesis made concrete: perceptual identity belongs in the parametric bank, where it scales, and exact facts belong in a text store, where lookup stays exact. The production path — perceptual bank for *who*, text store for *what* — sustains end-to-end task success at 0.83 for eighty users inside a single frozen model, with no external index.

6 The encoder determines recall

Grounding puts identity on a dedicated encoder, and that choice dominates everything downstream: because the read reproduces the encoder's recall, the memory's quality is the encoder's quality, and the design reduces to a few rules.

Choose the encoder well; nothing in-model recovers what it discards. The memory inherits the encoder's cross-condition invariance and nothing else, so recall moves with

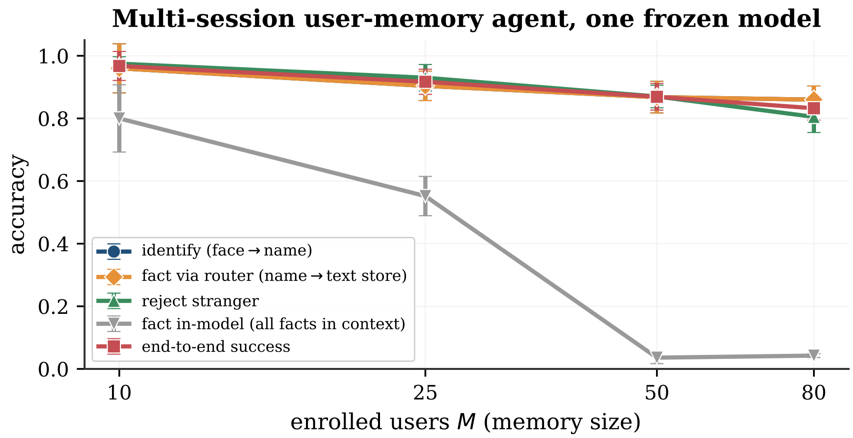


Figure 14. An end-to-end multi-session user-memory agent (Qwen2.5-3B, 5 seeds) as the enrolled population grows. Identifying a returning user (face→name), recalling their fact through the router (name→text store), and rejecting strangers all scale gracefully, tracking the encoder. Recalling the fact *in-model* instead, with all facts bound in context, collapses beyond a few dozen users — the host model's in-context limit, not the memory's — which is why the router keeps facts in a text store. End-to-end success holds at 0.83 for eighty users in one frozen model.

encoder quality on the *same* identities. On AgeDB, ArcFace reaches 0.95 recall at ten identities and 0.75 at a hundred, while a general vision-language model's own vision tokens manage only 0.64 and 0.34 on those exact faces; on LFW, ArcFace and AntelopeV2 agree to within two points. A learned metric on the same features does no better (Appendix A), and at large pools the encoder's own cosine is the best available. If a task needs better recall, the only remedy is a stronger encoder.

Use an external encoder for the key, not the model's native tokens. We run the memory end-to-end inside a real vision-language model (Qwen2.5-VL) that sees raw AgeDB face crops through its own visual front-end. The mechanism works — register one crop, recognise a cross-condition crop, all inside the VLM — but the recall is only the VLM's native-token recall, weak because those tokens are not face-specialised: 0.60 at ten identities, matching cosine over the same native keys (0.50) and far below a purpose-built encoder (Appendix Table 10). This is why the "identify" step uses a dedicated encoder and the VLM is reserved for the "localize" step it is actually good at: **the key should be a dedicated perceptual encoder, separate from the model's own representation**, which is also why the mechanism pairs naturally with an omni model's encoder used as an external feature rather than its in-context tokens.

The store is universal across frozen models, with one constant. The read ports to any frozen model without training (Appendix Figure 16, Table 5); the only model-dependent choice is the read-strength gain. On tied-embedding models the default suffices and the match to the encoder is exact on every draw; on untied models the marker rows are smaller in norm, so the gain is raised once, a constant set by inspection. The constant is forgiving rather than narrow: tied models are exact across the 64× range from 16 to 1024, untied models above a threshold (Mistral exact and stable from 256 through 1024, Appendix Table 6), and a non-transformer reader (hybrid-Mamba Granite) holds within a point. No family failed, and because nothing is trained, there is no convergence to be fragile.

7 Discussion

The claim is structural, and the numbers illustrate it. The argument is not "our method scores higher." It is that text is the wrong container for half of what an agent should remember about a person, that recognizing a perception decomposes into grounding it and identifying it, and that a frozen model fronted by a VLM and backed by an encoder bank solves both halves where text and a bare encoder each solve neither. The empirical results matter because they substantiate each piece: the text loss is real and modality-graded (Figure 6), the grounding step is necessary and lets the system recover the oracle (Figure 7), and the recognition core is the encoder's recall given an in-model home, training-free and universal (Appendix Figure 16). The contribution is the grounded, in-model container, not a recall score; the score is the encoder's.

This is additive infrastructure, not a replacement. The perceptual memory is designed to bolt onto the text memory that agents already have [Chhikara et al., 2025, Wang et al., 2024, Packer et al., 2023, Team, 2024], and the non-regression result is what makes that safe: text recall is preserved exactly, and the perceptual bank activates only at perceptual moments. As long-term memory benchmarks like LongMemEval [Wu et al., 2024] grow to include perceptual content, the gap this paper argues for becomes directly measurable.

The captionable half is going parametric too. This paper is one half of a larger bet: a personalized agent should keep its memory *inside* the model, not in an external index. A companion line makes the same bet for the *captionable* half. *User as Engram* [Li, 2026c] writes per-user *facts* into a hash-keyed parametric memory, showing that the obvious alternative (a per-user LoRA adapter) is *reasoning-negative*: it memorizes facts but degrades the reasoning that uses them, because a LoRA edit is global where a row insertion is local. *User as Code* [Li, 2026b] pushes the same half into executable typed state, and *Programmable KV Cache* [Li, 2026a] keeps it as editable, composable notes in the attention cache rather than in weights or an external index. Our perceptual counterpart shares that thesis with a different mechanism, continuous cross-attention over encoder banks rather than hash-keyed rows. That continuous channel, carrying a perception as a vector projected into the model's own embedding space rather than as text, is the same bet *The Latent Bridge* [Li and Shi, 2026] makes for inter-model communication, where a learned latent link beats writing and re-reading text. The two halves compose into one memory that holds both what the user said and what they are like.

The router: what a latent memory can and cannot hold. The split between the two stores is not a convention. It follows from two capacity laws that we measured by compressing content into k soft tokens read by a frozen language model (Figure 15). Perceptual identity is capacity-limited and degrades gracefully. Compressing M registered identities into k prototype slots and recognising a cross-condition query keeps a k/M fraction of them distinguishable, on faces, voices, acoustic scenes, painting styles, and tones of voice alike (Appendix Table 7). One slot per identity recognises as many as the encoder can, and fewer slots lose identities in proportion. The general law is recall $\approx \min(1, k/M) \cdot C(M)$, where $C(M)$ is the encoder's own recall at M registrations: near 1 for strong encoders (face, voice), lower for the weaker style and tone encoders, so the slot-compression factor $\min(1, k/M)$ is what is universal and the ceiling is the encoder's. This is not an artifact of k-means: a compressor whose k slots we instead learn by gradient descent lands on the same curve, because k hard slots can keep at most k of M identities separable (a pigeonhole bound). Exact factual content behaves in the opposite way. A latent holds one exact code and reads it back (0.87 for a six-character code), but storing several and retrieving one by name collapses at once (0.06 at two codes, near zero

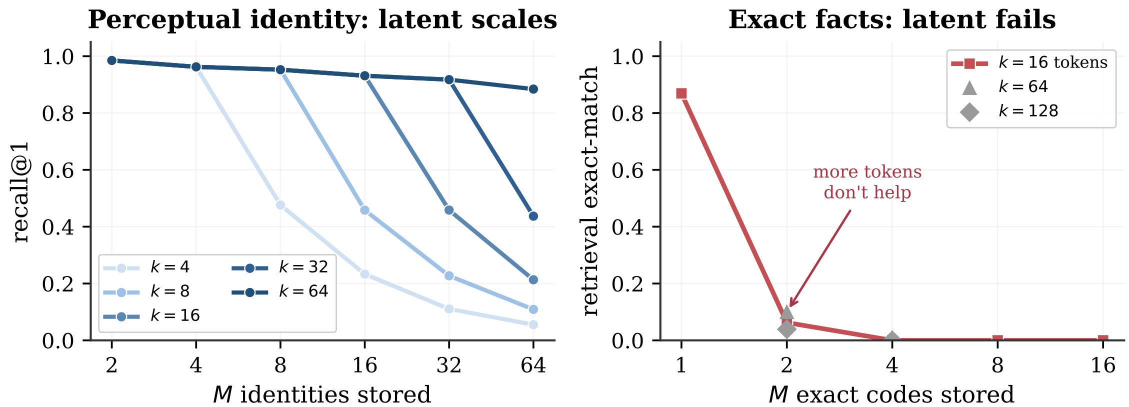


Figure 15. Two capacity laws for memory compressed into k soft tokens read by a frozen language model, the mechanism behind the router. **Left:** perceptual identity is capacity-limited. Compressing M faces into k prototype slots and recognising a cross-condition query gives recall $\approx \min(1, k/M)$, so a slot per identity recognises everyone and more slots hold more people (voices behave identically). **Right:** exact factual content is binding-limited. A latent holds one exact code (0.87) but multi-code retrieval collapses at two (0.06) and adding latent tokens does not rescue it. Recognition scales with the latent budget. Exact recall does not, so it belongs in a text store.

beyond), and adding latent tokens does not help. Recognition is a capacity problem, where a latent yields one identity per slot; exact recall is a binding problem, where a latent cannot match a key to its value at any budget. That is why the captionable half belongs in a text store, which performs that lookup exactly, and the perceptual half in a parametric bank, which scales with its slots — the same boundary the text ablation drew empirically (Figure 6), now derived.

Limitations. (i) The memory cannot exceed the encoder. Its recall *is* the encoder's, so at very large memories it declines exactly as the encoder's cosine does (0.77 at a thousand faces), and no in-model trick recovers more. (ii) We have measured real cross-condition data to roughly two thousand identities; the constant-time behaviour is confirmed to ten thousand, but recall at that scale awaits data. (iii) The grounding step inherits the grounder's referring-expression resolution, and it is the component that degrades with complexity: on well-separated visual composites it is near-lossless to six referents (grounding 1.00 through $K{=}5$), but photographic backgrounds with jitter drop it to 0.6–0.8, and audio grounding of the fourth speaker falls to 0.69; denser, occluded, or noisier scenes will stress it further, and detector or diarizer robustness is the practical bottleneck rather than the encoder. (iv) Inside a VLM, native vision tokens fail as keys and an external, modality-specific encoder is required. (v) The one per-model constant (read-strength gain) is set by inspection; untied-embedding models need a larger value, which we have not automated. (vi) The closest prior systems (MyVLM [Alaluf et al., 2024], Online-PVLM [Bai et al., 2025]) have released neither code nor checkpoints; we compare against charitable reimplementations of their core mechanisms (Appendix A, Table 9).

8 Conclusion

A photograph and a written description of a face are not two encodings of the same thing. The description has already thrown away what lets you recognize the person. Agent memory today keeps only the description; we have argued it must keep the photograph too, and that keeping it well means grounding it. A vision-language model resolves the referent the user means; a purpose-built encoder turns it into an identity that survives the conditions text cannot; and a frozen model stores that identity as a content-addressable row it reads inside its own forward pass. Each component does what the others cannot — the VLM cannot identify, the encoder cannot ground, and an external index cannot return the answer inside the model — and together they recover oracle-level recognition where each piece alone is near chance. The recognition itself is the encoder's, given a home: training-free, constant-time, and portable to any frozen model. The two memories are complementary, text for what the user *said* and parametric multimodal memory for what the user *is like*, and an agent with both strictly dominates one with only text. That is where a face stops being a sentence, and starts being a memory.

Acknowledgements

We thank Xiangyu Zhang from StepFun AI for the initial ideas that inspired this paper; he introduced the problem of multimodal user memory to the authors and pointed out the limitations of purely textual memory. This paper was produced using Pine Copilot's voice-directed *whisper coding* workflow [Pine AI, 2026], in which the authors specify, discuss, and review the work by voice while a coding agent (Claude Code with Claude Opus 4.8) carries out the planning, coding, experiments, and paper writing. We thank BSQL Networking for hosting the NVIDIA RTX PRO 6000 GPU.

A Detailed results

This appendix collects the per-cell numbers behind Sections 5–6. Recall is top-1 with the argmax restricted to registered markers.

Paired evaluation protocol. A comparison between the memory and retrieval requires two controls: both methods score on *identical* registrations and queries, and each target identity occupies a randomly chosen bank slot. The second control matters because a fixed target slot lets the constant marker token in that slot accrue a baseline output-logit, which inflates recall independently of the query. We score both methods on the same twenty draws with the target slot randomised, and report mean$\pm$std. Table 4 gives the result. The memory ties retrieval on random banks, since with a sharp read it computes the same argmax over encoder cosine, and on banks of the nineteen most-confusable look-alikes it trails by a few points: its similarity-weighted blend of marker values carries no per-identity signal beyond the cosine, so it cannot separate confusable identities the encoder already conflates.

Universality grid. Table 5 runs the training-free read across five model architectures and all five modalities, scoring each cell against the encoder with the same paired protocol (20 draws). Every one of the twenty-five cells reproduces the encoder's recall to within one point (worst case 0.010, most cells ≤ 0.003). Tied-embedding models use the default read-strength gain; untied and hybrid models use a single larger value (256), set once per model and held fixed across modalities. The read therefore inherits the encoder's recall regardless of the host model's family, embedding scheme, scale, or even whether it is a transformer.

Table 4. Paired evaluation: the memory and retrieval scored on identical registrations and queries across 20 draws, with the target slot randomised. The memory ties retrieval on random banks and trails it by a few points on look-alike banks. Recall is the encoder's throughout.

Regime	Cell	Retrieval	AttMem (mean±std)	Δ
random	Face, N=10	0.948	0.942 ± 0.035	-0.6pp
random	Style, N=5	0.473	0.476 ± 0.116	$+0.2$pp
random	Style, N=10	0.428	0.357 ± 0.081	-7.2pp
adversarial	Face, K=19	0.853	0.773 ± 0.008	-8.0pp

Table 5. Universality grid: maximum $|\Delta|$ between the training-free memory and the encoder over $N \in \{5, 10, 20\}$ (paired, 20 draws), for five architectures × five modalities. Every cell is within one point of the encoder. "gain" is the single per-model read-strength constant. Encoder recall@1 at N=10: Face 0.95, Speaker 0.99, Acoustic 0.86, Tone 0.52, Style 0.43.

Model	Emb.	gain	Face	Speaker	Acoustic	Tone	Style
Qwen2.5-7B	tied	64	0.001	0.000	0.002	0.003	0.001
Phi-3.5-mini	tied	64	0.001	0.000	0.000	0.002	0.002
DeepSeek-Llama-8B	untied	256	0.000	0.000	0.002	0.002	0.000
Mistral-7B	untied	256	0.000	0.000	0.003	0.010	0.000
Granite-4.0 (Mamba)	hybrid	256	0.003	0.000	0.003	0.001	0.000

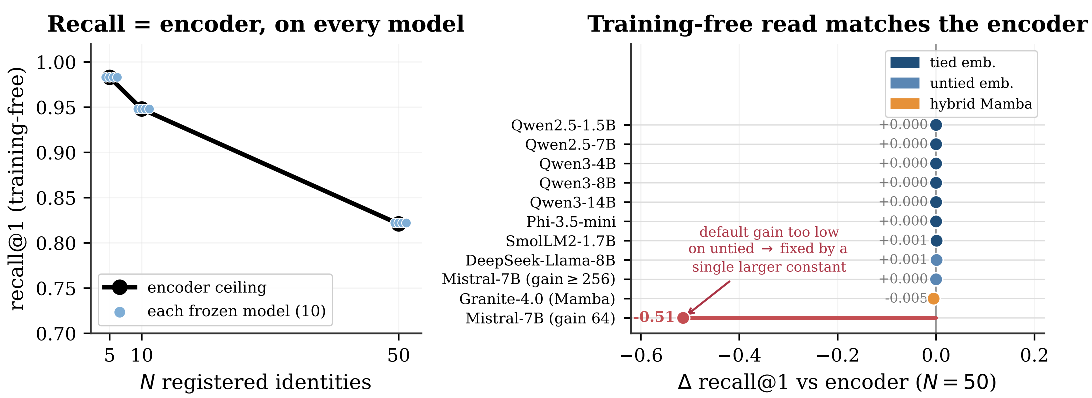


Figure 16. The training-free read reproduces the encoder on every frozen model, the supporting property behind the "store" step. **Left:** each model's training-free recall lands on the encoder ceiling (paired, 20 draws). **Right:** the gap from the encoder at N=50 is ≈ 0 across ten families (1.5B–14B, tied and untied embeddings, hybrid-Mamba); only an untied model at the default gain dips (red), fixed by a single larger read-strength constant. No gradient steps anywhere.

Read-strength robustness. The one per-model constant, the residual gain, is set by inspection rather than search because exact recall holds across a wide range (Table 6, face N=50, paired). A tied-embedding model is exact at every gain from 16 to 1024; an untied one (Mistral) is below the encoder until a threshold near 128–256 and then exact and stable through 1024. Any value in the upper range works for either, so there is no tuning to get wrong.

Capacity law across modalities. Table 7 runs the perceptual capacity probe (compress M registered identities into k prototype slots, recognise a cross-condition query, 5 seeds) on all five

Table 6. Read-strength robustness: training-free recall@1 (face, N=50, encoder ceiling 0.821) as the residual gain sweeps 16–1024. Tied models are exact throughout; untied models are exact above a threshold and stay exact. The constant tolerates a wide range.

Model	16	32	64	128	256	512	1024
Qwen2.5-3B (tied)	0.821	0.821	0.821	0.821	0.821	0.821	0.821
Mistral-7B (untied)	0.022	0.060	0.307	0.768	0.821	0.821	0.821

modalities. The slot-compression factor $\min(1, k/M)$ fits every modality once normalised by the one-slot-per-identity ceiling $C(M)$: the normalised error is ≤ 0.012 on the strong-encoder modalities and ≤ 0.13 on the weaker style and tone encoders. The ceiling $C(M)$ itself is the encoder's recall and varies accordingly, from near 1 on voice to 0.26–0.36 at M=32 on style and tone. Recognition is capacity-limited and graceful in every modality; only the ceiling moves.

Table 7. Perceptual capacity law per modality. $C(M)$ is the one-slot-per-identity ceiling (recall at k=M), i.e. the encoder's recall at M. "fit" is the mean $|\,\text{recall}/C(M) - \min(1, k/M)\,|$ over the k<M cells: the slot-compression factor is universal; the ceiling is the encoder's.

Modality	$C(2)$	$C(8)$	$C(32)$	**fit to** $\min(1, k/M)$
Face (ArcFace)	0.98	0.95	0.92	0.002
Voice (ECAPA)	1.00	1.00	0.99	0.001
Acoustic (AST)	0.95	0.87	0.76	0.012
Style (CLIP)	0.76	0.46	0.26	0.070
Tone (wav2vec)	0.94	0.67	0.36	0.125

Learned-metric baseline. A natural question is whether *any* re-encoding of the same features beats raw cosine, even if our read does not. We fit a similarity on the same encoder features and the same identities (the 50/50 split, seed 42) and score it with the identical protocol. Two closed-form metrics suffice: regularised LDA and within-class whitening. Table 8 shows they land on raw cosine at small memory and *below* it past a few hundred identities. So there is no discrimination left to extract from these features: the encoder's cosine is already the best ruler, which is why the memory's inheriting it exactly is the ceiling, not a shortfall. A contrastive linear head we also trained underperformed both closed-form metrics and is omitted as a weak instantiation.

Comparison with per-concept methods. The personalized-VLM line (MyVLM, Yo'LLaVA, Online-PVLM) registers a concept by *learning* something per concept (a classifier head, a soft token, a projection). Their code and checkpoints are unreleased, so we reimplement each method's core mechanism charitably on the cross-condition face task (ArcFace keys, same split). Table 9 reports accuracy and registration cost. Two findings. First, a per-concept linear classifier (MyVLM's mechanism) is statistically indistinguishable from raw cosine at every memory size, because with one registration embedding per identity the trained head collapses to the embedding itself. It yields no accuracy gain over retrieval, and it costs per-identity gradient descent that grows to 4.5 s at a thousand identities, against our $\mathcal{O}(1)$ tensor append. Second, a learned projection trained with supervised contrastive loss (Online-PVLM's mechanism) *underperforms* raw cosine by 15 to 28 points on this task. We read this as a caution that a

Table 8. Learned-metric baseline, recall@1, same encoder and same split as AttMem. A learned metric (LDA on faces, whitening on style) matches AttMem's small-N lift over raw cosine; at large N raw cosine wins for all. Style AttMem here is the seed-42 single run (the multi-seed mean is 0.64 at N=5); the comparison is within seed.

Modality	N	Raw cosine	LDA	Whitening	AttMem
Face (ArcFace)	5	0.933	**1.000**	0.933	0.933
Face (ArcFace)	10	0.933	0.967	0.933	**1.000**
Face (ArcFace)	300	0.734	0.667	**0.744**	0.641
Face (ArcFace)	1000	0.767	0.724	**0.776**	—
Style (CLIP-mid)	5	0.400	0.200	**0.600**	0.467
Style (CLIP-mid)	10	0.400	0.333	0.367	**0.467**

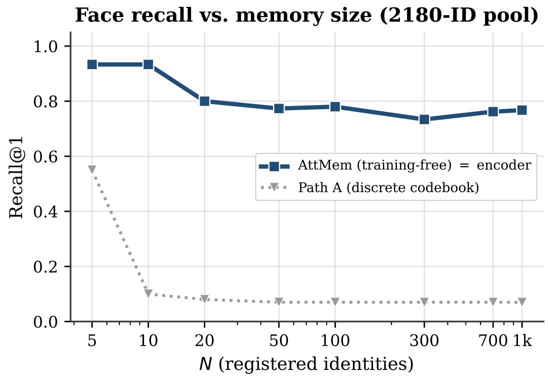
learned re-encoding is not without cost: tuned on identity-disjoint data, it can distort the very cross-condition geometry it was meant to sharpen. No method here, ours included, beats raw cosine on accuracy at scale; the parametric memory's case rests on cost and in-model behaviour, not recall.

Table 9. Charitable reimplementations of per-concept methods on cross-condition face recall (ArcFace, recall@1, single eval draw). MyVLM's per-concept classifier ties raw cosine but its registration cost grows with the memory; Online-PVLM's learned projection underperforms raw cosine. AttMem leads only at small N and trails at scale (the encoder-ceiling inversion).

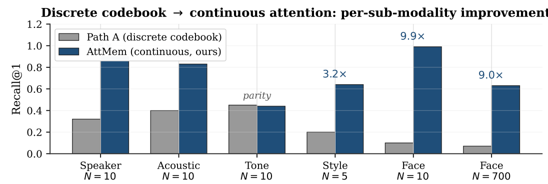
N	Raw cosine	MyVLM	Online-PVLM	AttMem	MyVLM reg. cost
5	0.933	0.933	0.800	0.933	0.009 s
10	1.000	0.967	0.750	0.992	0.019 s
50	0.767	0.753	0.675	0.733	0.178 s
100	0.780	0.770	—	0.742	0.528 s
300	0.728	0.725	—	0.637	1.39 s
1000	0.776	0.778	—	0.594	4.52 s

Scaling and the codebook comparison. Figure 17a traces recall against memory size on the face pool. The training-free memory tracks the encoder's own curve (it *is* that curve), declining gracefully from near-ceiling at ten identities to 0.77 at a thousand, while the discrete-codebook predecessor (PATH A) flatlines at $\sim$0.07 regardless of codebook size or budget. Figure 17b makes the same point across all five sub-modalities: replacing a discrete codebook with continuous attention lifts recall 2–10$\times$, because a categorical bottleneck discards the signal the encoder preserved. This is the one place a design choice in the read matters; everything else inherits the encoder.

Vision-language model, key/value orthogonality. Table 10 gives the numbers behind the structural rule of Section 6. Each memory row is compared to retrieval *in its own key space*. An external ArcFace key (orthogonal to the model's hidden space) reproduces the win. The VLM's native vision tokens (already in hidden space) only tie native-token retrieval and stay far below the ArcFace ceiling throughout, so they are the wrong key regardless of method — the empirical basis for reserving the VLM for localization and an external encoder for identity.



(a) scaling on the face pool



(b) codebook vs. continuous attention

Figure 17. Left: the continuous memory degrades gracefully with memory size (tracking the encoder ceiling), while the codebook predecessor never gets off the floor. Right: continuous attention beats the discrete codebook 2–10$\times$ on every sub-modality.

Table 10. Parametric memory inside Qwen2.5-VL-3B: same frozen model, only the key encoder changes. Recall at three memory sizes.

Configuration	N=10	N=100	N=1000
Retrieval over ArcFace embeddings (encoder ceiling)	0.933	0.780	0.767
AttMem + external ArcFace keys (orthogonal to hidden space)	**1.000**	0.710	0.494
Retrieval over native vision tokens	0.50	0.35	—
AttMem + native vision tokens (co-located with hidden space)	0.60	0.11	—

B Mechanics of a single recall

It helps to watch one recall end to end. Suppose the agent has met ten people and wants to register an eleventh from a single photo. Registration is one row appended to the face bank (Listing 1). The encoder turns the photo into a 512-dimensional ArcFace embedding, the *key*. The model's own embedding for a freshly assigned marker token becomes the *value*, a vector of the model's hidden width (2048 for Qwen2.5-3B). There is no gradient step and no fine-tuning. The row is the memory.

```
# Register identity #11 from a single photo -- no training:
k = l2_normalize(arcface(photo))        # key:   R^512  (encoder space)
m = assign_marker_token()               # e.g. "<id_11>"
v = model.input_embedding[m]            # value: R^2048 (model's own space)
bank.K = torch.cat([bank.K, k[None]])   # O(1) append
bank.V = torch.cat([bank.V, v[None]])
```

Listing 1. One registered identity is one row. The key is the encoder's view of the perception; the value is the model's own vector for the marker token that names it. Adding the row is a single tensor append.

At recall, a new photo of one of the eleven arrives under different lighting and a year older. Its ArcFace embedding becomes the query $\boldsymbol{q}$. Just before its output head, the model scores $\boldsymbol{q}$ against all eleven keys, softmaxes into attention weights $\boldsymbol{w}$, and pulls back the weighted blend of values $\boldsymbol{r} = \boldsymbol{w}^\top \boldsymbol{V}$. Because each value is the model's own vector for a marker token, adding

$g\,W_o\boldsymbol{r}$ to the hidden state lands as a clean boost on the right marker's logit. The model's next token *is* the remembered identity. The whole step is a few matrix multiplies dwarfed by the model's forward pass.

The 0.98-vs-0.46 diagonal gap of Section 5 (Figure 12) is measured on exactly this probe: a held-out ten-identity bank, comparing the sharp attention weights against the raw encoder cosine they are computed from. The attention concentrates the cosine onto its top match, which is why the read recovers the encoder's nearest neighbour as a single decisive marker rather than a diffuse blend.

C A discrete-codebook alternative

Before continuous attention we spent sixteen development cycles on PATH A: a discrete codebook that snaps each perception to one of K learned codes and remembers the code. It is the natural "compress the perception into a symbol" design, and it is a learned cousin of captioning, which is why its failure is instructive rather than embarrassing.

No setting broke it. Across codebook sizes $K \in \{128, 256, 512, 1024\}$, even after *100,000* steps of continual pretraining on the expanded face pool, top-1 recall at 300 identities stays pinned between 0.057 and 0.070 while embedding retrieval over the same encoder reaches 0.73. That is a roughly $10\times$ gap that does not move with K (Table 11). Swapping ArcFace R50 for AntelopeV2 R100 trained on 360K identities did not help either.

Table 11. PATH A after 100K-step continual pretraining on the 2180-identity face pool, at 300 registered identities. Recall is pinned near 0.07 regardless of codebook size, even when the gate routes to the correct code about half the time.

Codebook size K	128	256	512	1024
PATH A recall@1	0.057	0.064	0.070	0.070
gate routes to correct code	0.43	0.51	0.42	0.54
embedding retrieval (same encoder)		0.73		

The reason is a vice the codebook cannot escape, and the diagnostics name it exactly. Two pressures pull in opposite directions as K grows. A *small* codebook packs many people into each cell: at K=16, distinct identities collide in the same code 8.7% of the time, and once two people share a code they are indistinguishable forever after. A *large* codebook fixes that, with inter-identity collisions falling to 2.4% at K=128, but it shatters each identity across cells. The rate at which two cross-condition photos of the *same* person land in the same code drops from 0.33 to 0.20, so the query no longer routes to where the registration was stored. Net recall is squeezed from both sides and never escapes $\sim$0.07. Even when the gate does route a query to the right code (about half the time, Table 11), every other identity sharing that cell is an equally good answer, so the final pick is near-random.

This is the same information loss as captioning, learned instead of written: any categorical bottleneck, whether a word or a code, throws away the continuous signal the encoder worked to preserve, and no amount of compute downstream can recover it. Continuous attention over the raw embedding never quantizes, and never pays this tax. PATH A is in the paper because its failure is the cleanest possible argument for the design that replaced it.

References

Yuval Alaluf, Elad Richardson, Sergey Tulyakov, Kfir Aberman, and Daniel Cohen-Or. MyVLM: Personalizing VLMs for user-specific queries. In *European Conference on Computer Vision (ECCV)*, 2024.

Ruichuan An, Sihan Yang, Ming Lu, Kai Zeng, Yulin Luo, Ying Chen, Jiajun Yong, Huanqia Liang, Qi Huang, and Shanghang Zhang. MC-LLaVA: Multi-concept personalized vision-language model. *arXiv preprint arXiv:2411.11706*, 2024.

Alexei Baevski, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in Neural Information Processing Systems*, 2020.

Huiyu Bai, Runze Wang, Zhuoyun Du, Yiyang Zhao, Fengji Zhang, Haoyu Chen, Xiaoyong Zhu, Bo Zheng, and Xuejiao Zhao. Online-PVLM: Advancing personalized VLMs with online concept learning. *arXiv preprint arXiv:2511.20056*, 2025.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready AI agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025.

Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. ArcFace: Additive angular margin loss for deep face recognition. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification. In *INTERSPEECH*, 2020.

Yuan Gong, Yu-An Chung, and James Glass. AST: Audio spectrogram transformer. In *INTERSPEECH*, 2021.

Haoran Hao, Jiaming Han, Changsheng Li, Yu-Feng Li, and Xiangyu Yue. RAP: Retrieval-augmented personalization for multimodal large language models. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

Gary B Huang, Marwan Mattar, Tamara Berg, and Eric Learned-Miller. Labeled faces in the wild: A database for studying face recognition in unconstrained environments. In *Workshop on Faces in Real-Life Images*, 2008.

Urvashi Khandelwal, Omer Levy, Dan Jurafsky, Luke Zettlemoyer, and Mike Lewis. Generalization through memorization: Nearest neighbor language models. In *International Conference on Learning Representations (ICLR)*, 2020.

Bojie Li. Models take notes at prefill: KV cache can be editable and composable. *arXiv preprint arXiv:2606.17107*, 2026a.

Bojie Li. User as code: Executable memory for personalized agents. *arXiv preprint arXiv:2606.16707*, 2026b.

Bojie Li. User as engram: Internalizing per-user memory as local parametric edits. *arXiv preprint arXiv:2606.19172*, 2026c.

Bojie Li and Noah Shi. The latent bridge: A continuous slow–fast channel for real-time game agents. *arXiv preprint arXiv:2606.24470*, 2026.

Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning (ICML)*, 2023.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Steven R Livingstone and Frank A Russo. The Ryerson audio-visual database of emotional speech and song (RAVDESS). volume 13, 2018.

Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory (M3-Agent). *arXiv preprint arXiv:2508.09736*, 2025.

Stylianos Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. AgeDB: The first manually collected, in-the-wild age database. In *CVPR Workshops*, 2017.

Thao Nguyen, Haotian Liu, Yuheng Li, Mu Yi, Jiarui Mu, and Yong Jae Lee. Yo'LLaVA: Your personalized language and vision assistant. *arXiv preprint arXiv:2406.09400*, 2024.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. LibriSpeech: An ASR corpus based on public domain audio books. 2015.

Karol J Piczak. ESC: Dataset for environmental sound classification. In *ACM Multimedia*, 2015.

Pine AI. Pine AI: The most natural human-computer interface is your voice. Blog post, 2026. URL https://www.19pine.ai/blog/pine-ai-the-most-natural-human-computer-interface-is-your-voice. Accessed 2026-06-28.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*, 2022.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-Networks. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

Letta Team. Letta (formerly MemGPT): Stateful LLM agents with long-term memory. *arXiv preprint arXiv:2310.08560*, 2024. Software framework.

Yu Wang, Dmitry Krotov, Yuanzhe Hu, Yifan Gao, Wangchunshu Zhou, Julian McAuley, Dan Gutfreund, Rogerio Feris, and Zexue He. MemoryLLM: Towards self-updatable large language models. *arXiv preprint arXiv:2402.04624*, 2024.

WikiArt. WikiArt: Visual art encyclopedia. 2010. https://www.wikiart.org.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*, 2024.

Yuhuai Wu, Markus N Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations (ICLR)*, 2022.